Bayesian Wave Inversion and Uncertainty Quantification

Markov Chain Monte Carlo for Parameter Estimation in the Wave Equation

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Mathematical Foundations of Computational Science

Motivation and aim

Waveform inversion: infer an inaccessible medium $c(x)$ from waves emitted by sources and recorded at receivers.

- Measurements are noisy $\Rightarrow$ the inverse problem is ill-posed
- A single best estimate is not enough: how certain are we?

We parameterize a localized contrast by two unknowns,

$$\rho(x) = a \exp\left(-\frac{(x - x_0)^4}{2w^2}\right), \qquad \theta = (a, w) \;\text{ or }\; \theta = (a, p),\; x_0 = x_0(p).$$

Aim

1. **Invert** the parameters θ from noisy scattered wave data
2. **Quantify the uncertainty**: posterior distribution, credible intervals, parameter coupling
3. Understand how the **noise level** controls this uncertainty

1 Model and data

2 Bayesian inference

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Wave model and ROM linearization

A pulse f emitted at x_s probes a medium with wave speed $c(x)$:

$$\left[\partial_t^2 - c^2(x)\Delta\right] p^{(s)}(t,x) = f(t)\,\delta_{x_s}(x)$$

With $\mathcal{A}_c = -c\,\Delta(c\,\cdot)$, the data at receiver x_r read

$$D_{r,s}(t) = \int_\Omega u_0^{(r)}(x)\,u^{(s)}(t,x)\,dx, \qquad u^{(s)}(t,\cdot) = \cos\!\left(t\sqrt{\mathcal{A}_c}\right) u_0^{(s)}$$

Scattered data relative to the background c_0 (Lippmann–Schwinger):

$$D^{\mathrm{sc}}(t) = \int_0^t \int_\Omega \rho(x)\,u^{(s)}(t',x)\,\partial_t u^{(r)}(t-t',x;c_0)\,dx\,dt', \qquad \rho = \frac{c^2 - c_0^2}{c\,c_0}$$

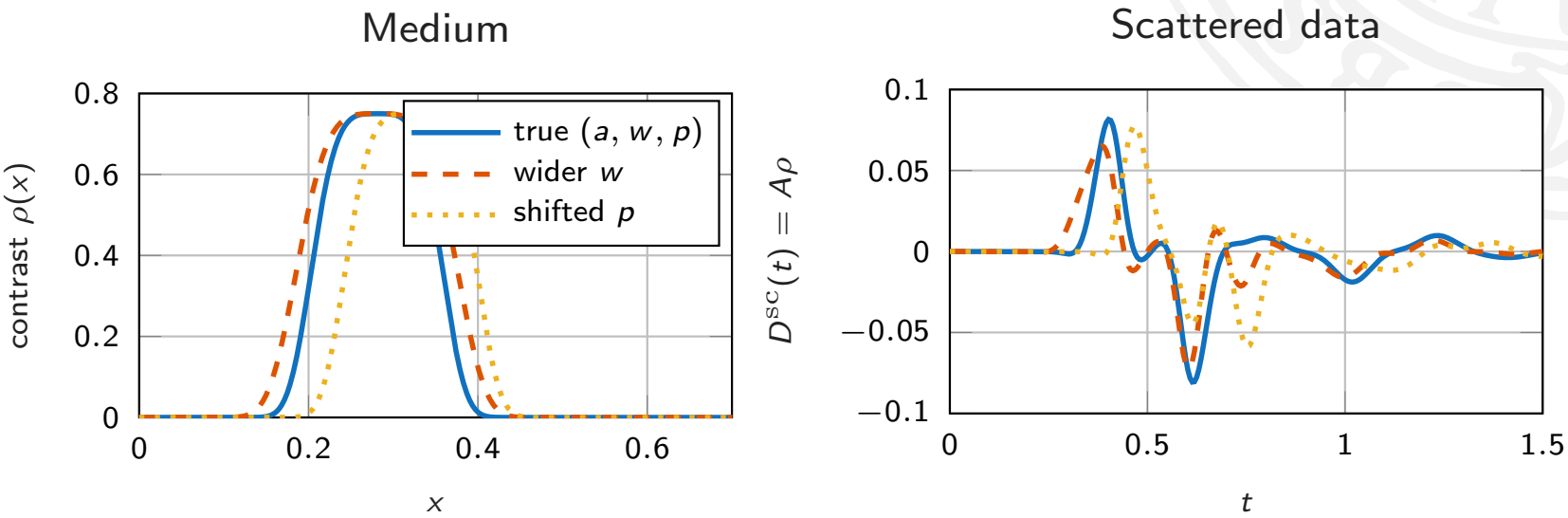
ROM linearization

Nonlinear in c, since $u^{(s)}$ depends on c. A data-driven reduced-order model [1] provides $u^{(s)}$

$\Rightarrow$ the forward map becomes **linear** in the contrast: $\quad D^{\mathrm{sc}} = A\rho$

Linear forward map and data

$$\theta \longmapsto \rho_\theta(x) = a \exp\left(-\frac{(x - x_0)^4}{2w^2}\right) \longmapsto D^{\mathrm{SC}}(t;\theta) = A\rho_\theta$$



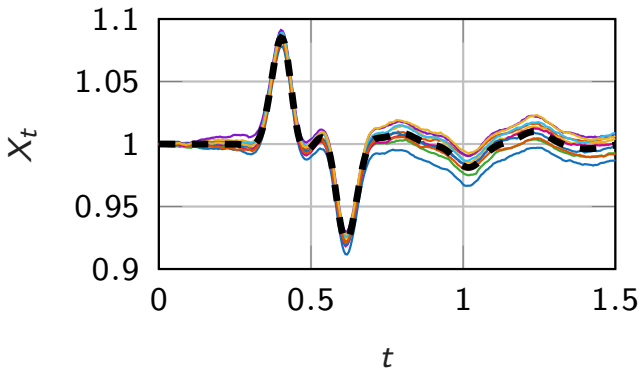
- Amplitude a: scales the data (linear)
- Width w: changes the shape of the reflections
- Position p: shifts the arrival times

Noisy measurements

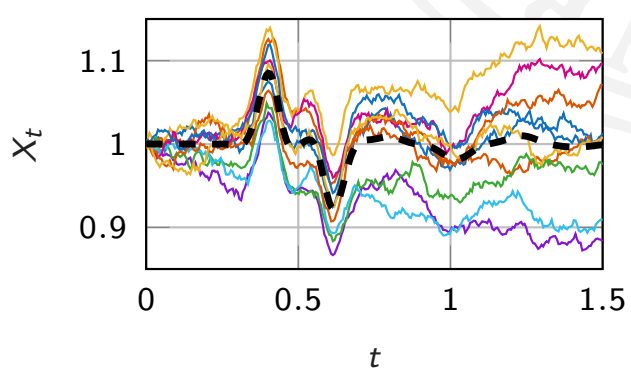
Multiplicative noise driven by a Brownian motion W_t; Gaussian log-increments ($Y_k = \log X_{t_k}$):

$$X_t = X_0 \exp\big(D^{\mathrm{SC}}(t;\theta) + \sigma W_t\big), \qquad \Delta Y_k = \Delta D_k^{\mathrm{SC}}(\theta) + \sigma\sqrt{\Delta t}\,\xi_k,\ \ \xi_k \sim \mathcal{N}(0,1)$$

$\sigma = 0.005$: 17% per path, 1.7% effective



$\sigma = 0.05$: 170% per path, 17% effective



Ten of $M = 100$ observed trajectories (solid); noise-free signal (dashed).

Relative noise level

$\|\sigma W\|/\|D^{\mathrm{SC}}\| \approx 34\,\sigma$ per trajectory. The $M = 100$ independent trajectories act like an average and divide the noise by $\sqrt{M} = 10$: **effective relative noise** $\approx 3.4\,\sigma$ (e.g. $\sigma = 0.1 \to 34\%$).

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Bayesian formulation

Log-likelihood of the M observed trajectories:

$$\log L(\theta) = -\frac{1}{2\sigma^2 \Delta t} \sum_{m=1}^{M} \sum_{k=0}^{N-1} \left(\Delta Y_k^{(m)} - \Delta D_k^{\mathrm{sc}}(\theta) \right)^2 + \text{const}$$

Posterior (Bayes) [7], with a flat prior $\pi(\theta) = \text{const}$:

$$\pi(\theta \mid \mathcal{D}) = \frac{L(\theta)\,\pi(\theta)}{Z} \propto L(\theta), \qquad Z = \int_{\Theta} L(\theta)\,\pi(\theta)\,d\theta$$

- The MLE $\hat{\theta} = \arg\max L(\theta)$ gives a point estimate only
- Z is intractable $\Rightarrow$ sample the posterior with MCMC, using only the unnormalized density
- From the samples: posterior mean, credible intervals, covariance, correlation

Metropolis–Hastings algorithm

Random-walk Metropolis–Hastings

Start from $\theta^{(0)}$. For $n = 0, 1, 2, \ldots$

1. **Propose** $\theta' = \theta^{(n)} + \varepsilon, \quad \varepsilon \sim \mathcal{N}\big(0, \mathrm{diag}(s_a^2, s_\lambda^2)\big)$
2. **Accept** with probability

$$\alpha = \min\left(1,\ \frac{\pi(\theta' \mid \mathcal{D})}{\pi(\theta^{(n)} \mid \mathcal{D})}\right) = \min\left(1,\ e^{\log L(\theta') - \log L(\theta^{(n)})}\right)$$

3. If accepted, $\theta^{(n+1)} = \theta'$; otherwise $\theta^{(n+1)} = \theta^{(n)}$

- Symmetric proposal $\Rightarrow$ the proposal densities cancel in α
- Detailed balance $\Rightarrow$ the chain converges to $\pi(\theta \mid \mathcal{D})$ [4, 5]
- Only $\log L$ is needed: one forward solve $A\rho(\theta')$ per iteration
- Tuning parameters: the step sizes s_a, s_λ (see proposal scaling)

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Laplace approximation: covariance and correlation

Second-order expansion of $\log L$ around $\hat{\theta}$ gives a Gaussian posterior [7]:

$$\pi(\theta \mid \mathcal{D}) \approx \mathcal{N}\left(\hat{\theta},\ \sigma^2 G^{-1}\right), \qquad G = \frac{M}{\Delta t} \sum_{k=0}^{N-1} \nabla_\theta \Delta D_k^{\mathrm{sc}} \, \nabla_\theta \Delta D_k^{\mathrm{sc}\,\top}$$

G depends only on the **forward model**, not on the noise.

Depend on the noise

$$\mathrm{Var}(\theta_i) \approx \sigma^2 \, (G^{-1})_{ii}$$

$$\mathrm{Cov}(a, \lambda) \approx -\sigma^2 \, \frac{G_{12}}{\det G}$$

Both grow like σ^2: growing covariance $\neq$ stronger coupling.

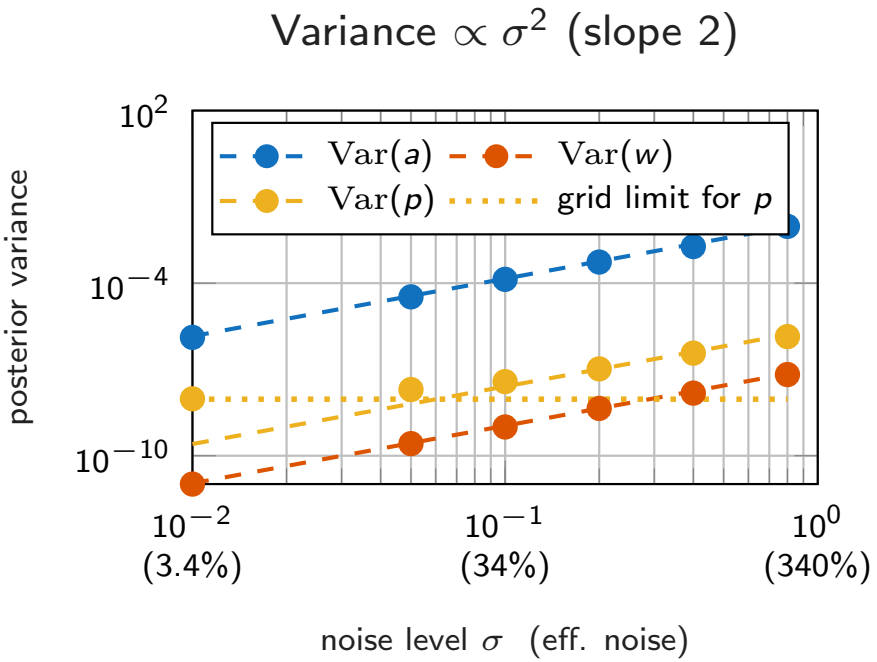
Independent of the noise

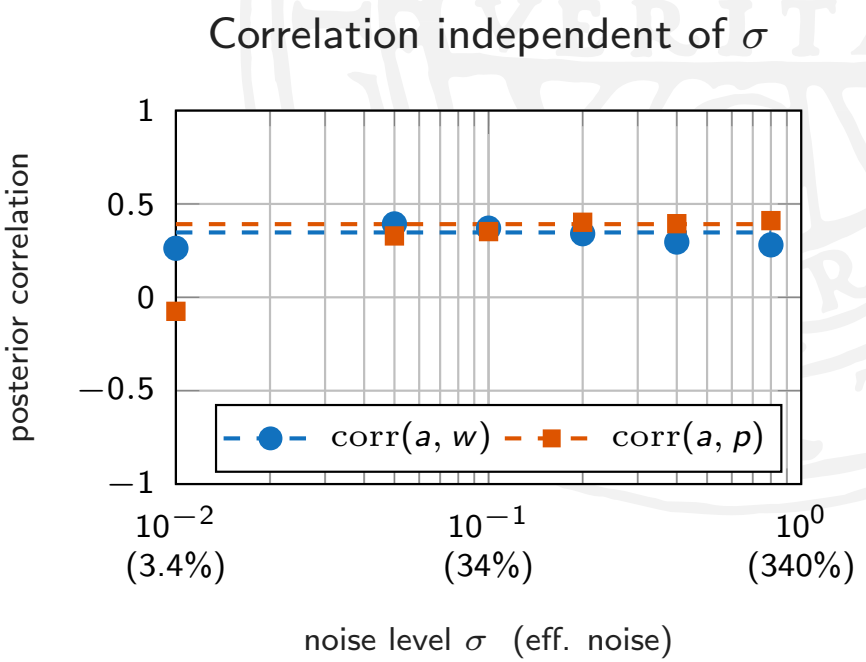
$$\mathrm{corr}(a, \lambda) = \frac{\mathrm{Cov}(a, \lambda)}{\sqrt{\mathrm{Var}(a)\,\mathrm{Var}(\lambda)}}$$

$$\mathrm{corr}(a, \lambda) \approx -\frac{G_{12}}{\sqrt{G_{11}\, G_{22}}}$$

σ^2 cancels: the coupling is fixed by the forward model.

Posterior variance grows quadratically with the noise





- MCMC (markers) follows the Laplace prediction (dashed): fitted slopes 2.01 for Var(a) and 1.99 for Var(w)
- Var(p) levels off at low noise: the position is resolved only up to one grid cell
- Correlations stay at corr(a, w) ≈ 0.35 and corr(a, p) ≈ 0.39 for all σ

Proposal scaling

The step sizes decide whether the chain explores the posterior efficiently:

- steps **too large** $\Rightarrow$ proposals land in low-probability regions, almost all rejected, the chain gets stuck
- steps **too small** $\Rightarrow$ almost all accepted, but the chain moves slowly

Scale the proposal to the posterior width

$$s_i = \frac{2.4}{\sqrt{d}}\,\sigma\sqrt{(G^{-1})_{ii}} \overset{d=2}{\approx} 1.7\,\sigma\sqrt{(G^{-1})_{ii}}, \qquad i \in \{a, \lambda\}$$

- $\sigma\sqrt{(G^{-1})_{ii}}$ is the Laplace posterior standard deviation
- The factor $2.4/\sqrt{d}$ is the optimal random-walk scaling for a Gaussian target [3, 6], giving acceptance rates of about 20–40%
- The steps must shrink with the noise level, since the posterior width is proportional to σ

1 Model and data

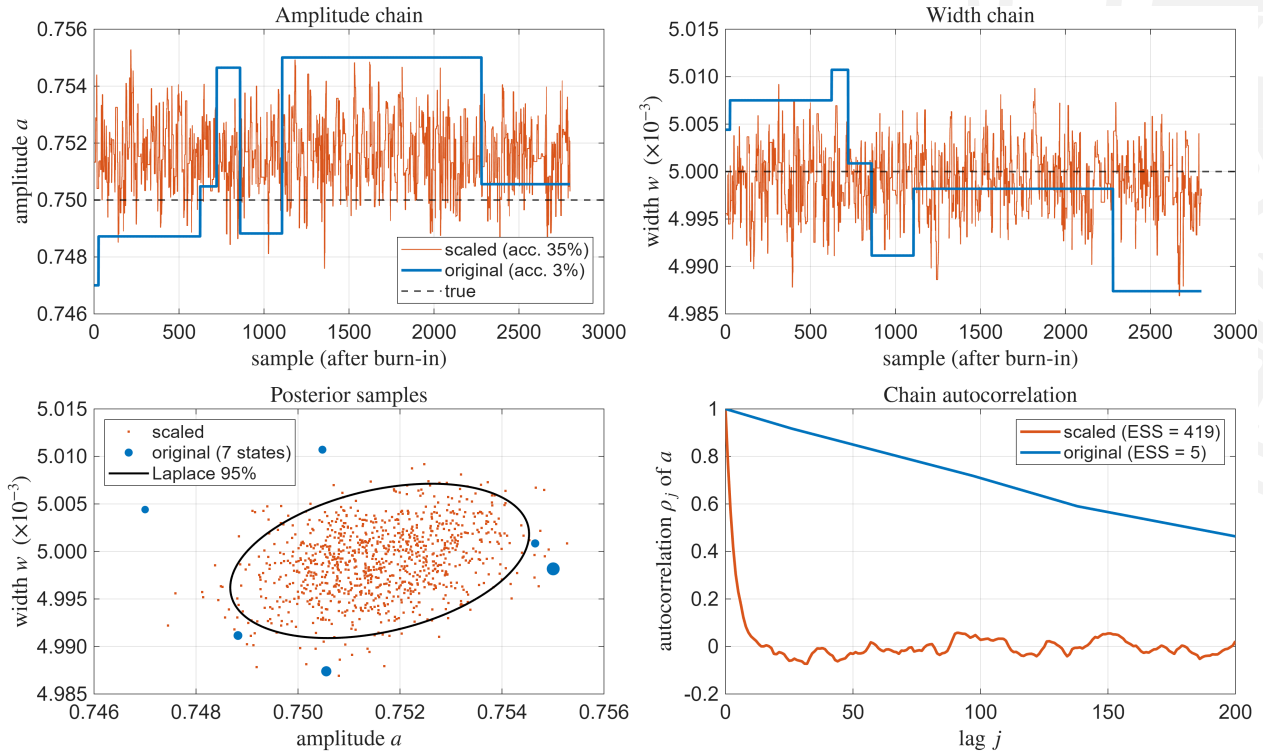
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Proposal scaling in practice



- (a, w), $\sigma = 0.01$, same data, 2800 samples each; posterior std $\approx (1.2 \cdot 10^{-3}, 3.3 \cdot 10^{-6})$
- **Original** steps $(10^{-2}, 10^{-3})$, $\sim$300$\times$ too wide in w: 3% accepted, 7 states, none in the 95% region
- **Scaled** steps $(2.0 \cdot 10^{-3}, 5.6 \cdot 10^{-6})$: 35% accepted, 94% of the samples in the 95% region

Proposal scaling and the accuracy of some estimates

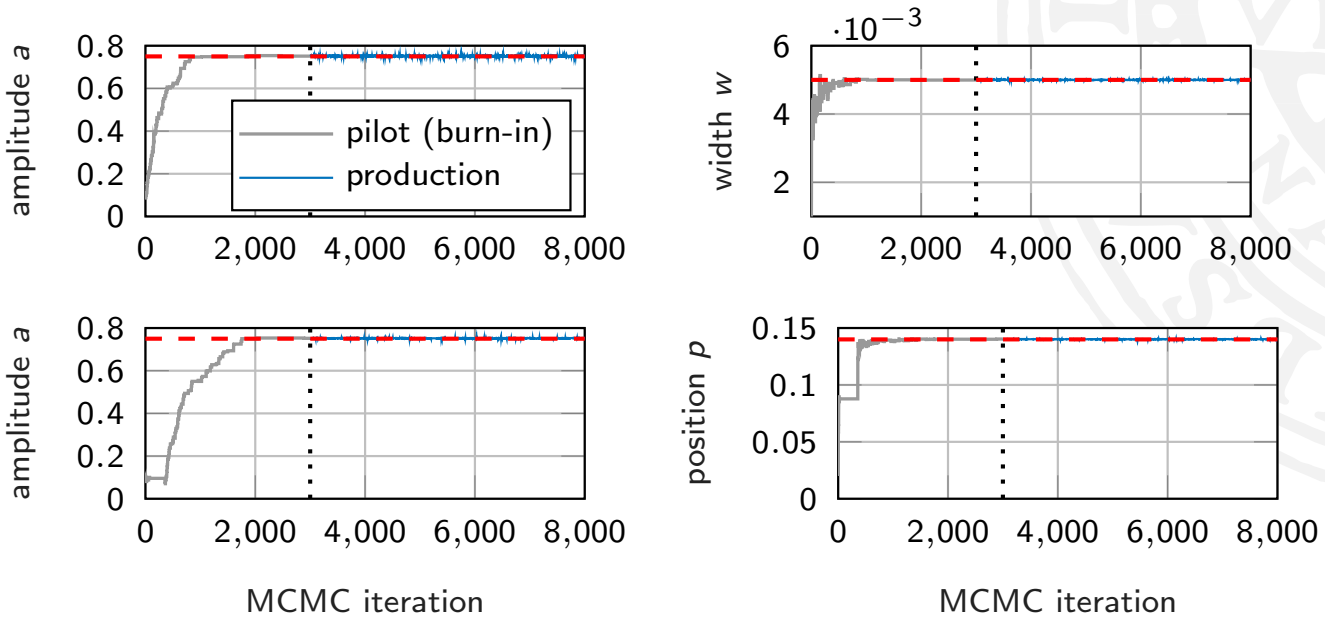
Effective sample size $N_{\text{eff}} = N_s/\tau$, $\tau = 1 + 2\sum_{j\geq 1}\rho_j$: Monte Carlo error of a posterior mean $\approx \text{std}/\sqrt{N_{\text{eff}}}$.

	original proposal				scaled proposal			
σ	acc.	$N_{\text{eff}}(a)$	$\text{std}(a)/\text{std}_{\text{Lap}}$	$\text{corr}(a,w)$	acc.	$N_{\text{eff}}(a)$	$\text{std}(a)/\text{std}_{\text{Lap}}$	$\text{corr}(a,w)$
0.01	2.6%	5	2.37	−0.11	34.9%	419	0.99	0.35
0.05	3.1%	27	0.97	0.25	33.3%	327	1.00	0.32
0.1	4.9%	19	0.81	0.16	33.9%	387	0.97	0.33
0.2	8.5%	8	1.09	0.48	34.1%	283	0.97	0.36
0.4	15.1%	10	0.69	0.17	31.8%	380	0.97	0.37
0.8	31.0%	11	0.64	0.22	34.7%	387	1.02	0.24

(a, w), 2800 samples per chain. Reference: Laplace std_{Lap} and $\text{corr}(a, w) = 0.35$.

- **Original**: $N_{\text{eff}} = 5$–27; std wrong by -36% to $+137\%$; correlation between -0.11 and 0.48
- **Scaled**: $N_{\text{eff}} \approx 300$–$400$; std within 3%; correlation close to 0.35
- Same cost (one forward solve per iteration), about $9\times$ smaller Monte Carlo error at $\sigma = 0.01$

Successful inversion: the sampled Markov chains



- $\sigma = 0.01$; start far from the truth: $(a, w) = (0.1,\ 10^{-3})$, $(a, p) = (0.1,\ 0.02)$
- **Pilot** (grey, fixed steps): reaches the true values (red) in 1000–2000 iterations
- **Production** (blue): steps from G at the best pilot state; acceptance 35% and 19%

How the uncertainty is quantified

From the production samples $\theta^{(n)}$, $n = 1, \ldots, N_s$, of $\pi(\theta \mid \mathcal{D})$:

Parameters

- Estimate: posterior mean $\bar{\theta} = \frac{1}{N_s} \sum_{n=1}^{N_s} \theta^{(n)}$
- 95% credible interval: $[\, q_{0.025},\ q_{0.975} \,]$, empirical quantiles of $\{\theta_i^{(n)}\}_n$, i.e. θ_i lies in it with posterior probability 0.95

Medium (uncertainty region)

- Map every sample to its wave-speed profile $c^{(n)}/c_0 = \frac{1}{2}\left(\rho_{\theta^{(n)}} + \sqrt{\rho^2_{\theta^{(n)}} + 4}\,\right)$
- **Inferred medium**: pointwise mean over n
- **Uncertainty region**: pointwise 2.5% and 97.5% quantiles over n, at each x

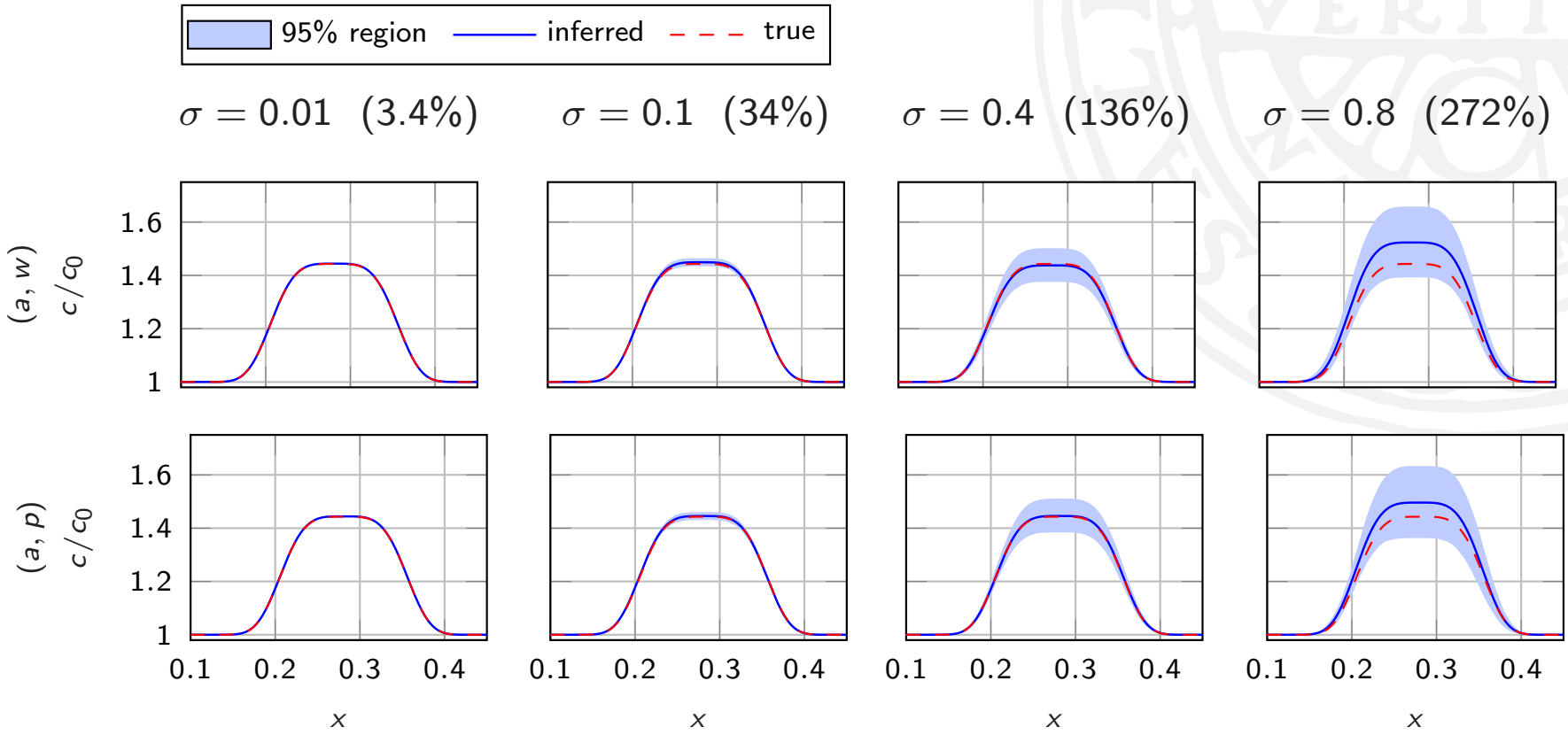
Parameter estimates

Posterior mean and 95% credible interval from the 5000 production samples:

problem	parameter	true	$\sigma = 0.01$ (3.4% noise)	$\sigma = 0.8$ (272% noise)
(a, w)	a	0.75	0.7516 [0.7493, 0.7539]	0.865 [0.675, 1.051]
	w $(\times 10^{-3})$	5	4.999 [4.992, 5.006]	5.11 [4.63, 5.69]
(a, p)	a	0.75	0.7518 [0.7496, 0.7541]	0.826 [0.626, 1.030]
	p	0.14	0.14016 [0.14001, 0.14033]	0.1395 [0.1370, 0.1420]

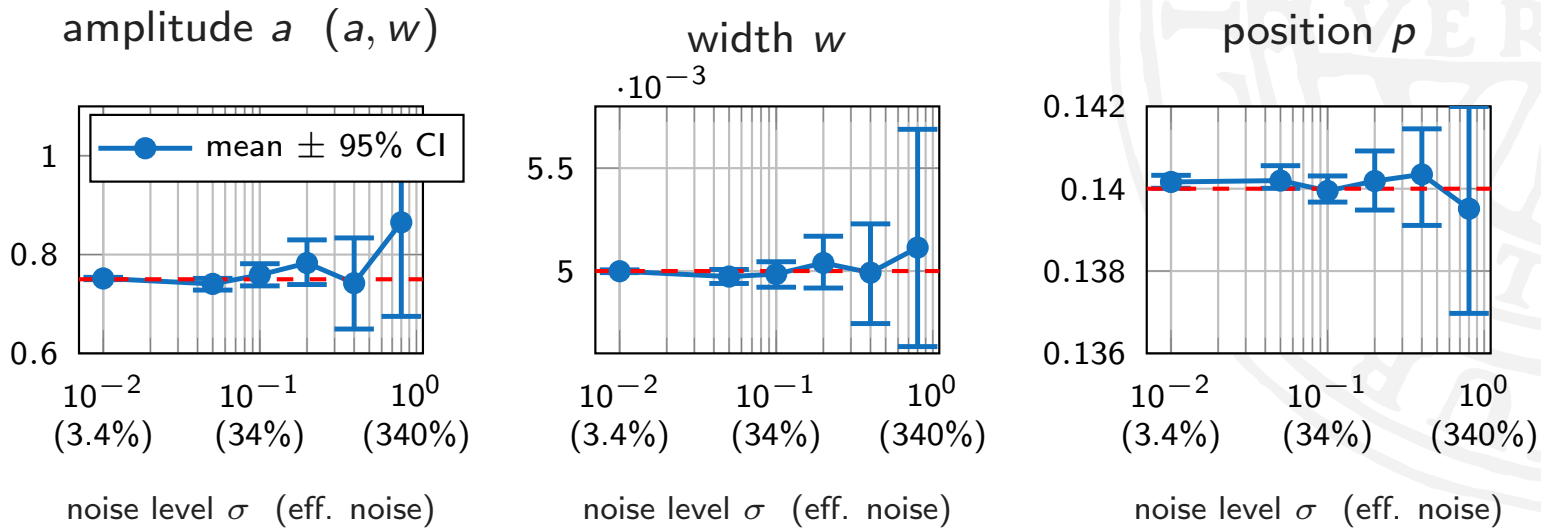
- Low noise: relative errors below 0.3%, very narrow intervals
- High noise: much wider intervals, but they still contain the true values
- The intervals contain the true values at **all six** noise levels, in both problems

Uncertainty region of the reconstructed medium



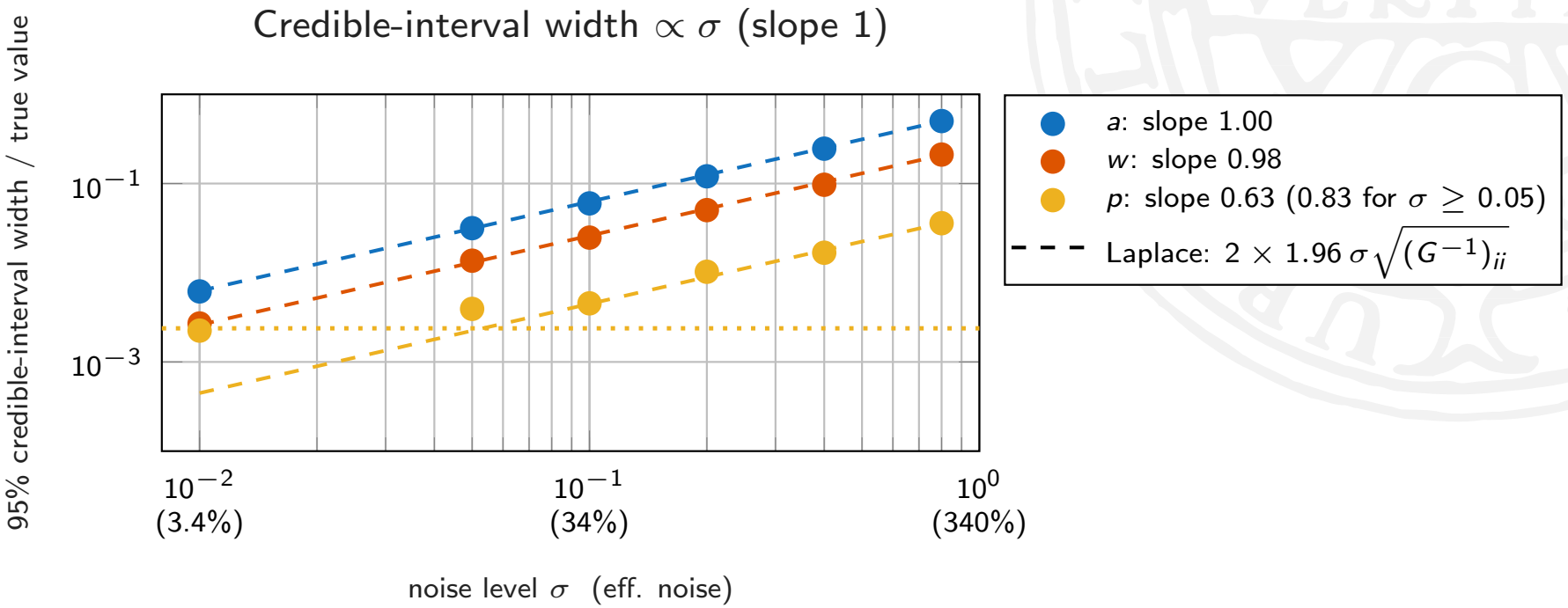
- The 95% region widens with the noise, mostly at the top: the **amplitude** is the least certain
- At $\sigma = 0.8$ the inferred medium is biased, but the true medium stays inside the region

Credible intervals versus noise



- Axis: σ and, in brackets, the effective relative noise of the $M = 100$ trajectories
- Interval widths grow in proportion to σ: log-log slopes 1.00 (a), 0.98 (w), 0.83 (p, $\sigma \geq 0.05$) (next slide)
- The true values (red) lie inside every 95% interval
- At large noise the posterior mean drifts from the truth: the posterior is no longer Gaussian

Credible-interval width grows linearly with the noise



- Width $= q_{0.975} - q_{0.025}$ over the true value; slope = least-squares fit of log width vs $\log \sigma$
- a, w: slopes 1.00, 0.98, on the Laplace lines $\Rightarrow$ width $\propto \sigma$
- p: 0.83 for $\sigma \geq 0.05$; at low noise limited by one grid cell (dotted)

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Take-aways

1. **Inversion works even with fixed steps**: from a far initial guess the chains reach the true profile.
   But they locate the posterior rather than sample it.
2. **Bayesian inversion quantifies the uncertainty**: the noise model enters the likelihood explicitly; the posterior gives credible intervals and parameter coupling, while deterministic inversion returns a single estimate.
   Proposal scaling is needed for accurate sampling and UQ: N_{eff} 5 $\rightarrow$ 400; std error up to 137% $\rightarrow$ 3%; correct credible intervals and correlations.
3. **The acceptance rate is a first check, not a proof**: at $\sigma = 0.8$ the fixed steps accept 31%, yet $N_{\text{eff}}(a) = 11$
   (the a-steps are 10$\times$ too small). Check per-parameter ESS and autocorrelation.
4. **The sign of the correlation reveals the physics**:
   - if only the L^1 norm $\int \rho \, dx \propto a\sqrt{w}$ scattered: exact trade-off, $\text{corr}(a, w) \approx -1$
   - observed $\text{corr}(a, w) = +0.35$ (MCMC and Laplace $-G_{12}/\sqrt{G_{11}G_{22}}$)
   - reflections come from the contrast *gradients*: a wider, smoother bump reflects less and needs a larger amplitude

Outlook

How many parameters can we invert?

- Here: 2 parameters of a bump of *known shape*: 0.3% error at 3.4% noise
- General media: expand $\rho = \sum_j \beta_j \phi_j$ with a prior. How much information do the data contain?

Beyond the L^2 misfit: Wasserstein distance

- L^2 compares signals point by point in time: shifted arrivals cause cycle skipping $\Rightarrow$ non-convex in position, steep jumps in the position chain
- Signals $\to$ probability densities on $[0, T]$: $f = e^{c\,\bar{Y}} / \int_0^T e^{c\,\bar{Y}}\,dt$ (data), g_θ likewise from $D^{\mathrm{sc}}(\cdot;\theta)$; CDF $F(t) = \int_0^t f$
- Optimal transport [2]: least squared distance to move the mass of f onto g; in 1-D the optimal map is monotone, $\mathcal{T} = G^{-1} \circ F$:

$$W_2^2(f,g) = \min_{\mathcal{T}_\# f = g} \int_0^T |t - \mathcal{T}(t)|^2 f(t)\,dt = \int_0^1 \left|F^{-1}(u) - G^{-1}(u)\right|^2 du$$

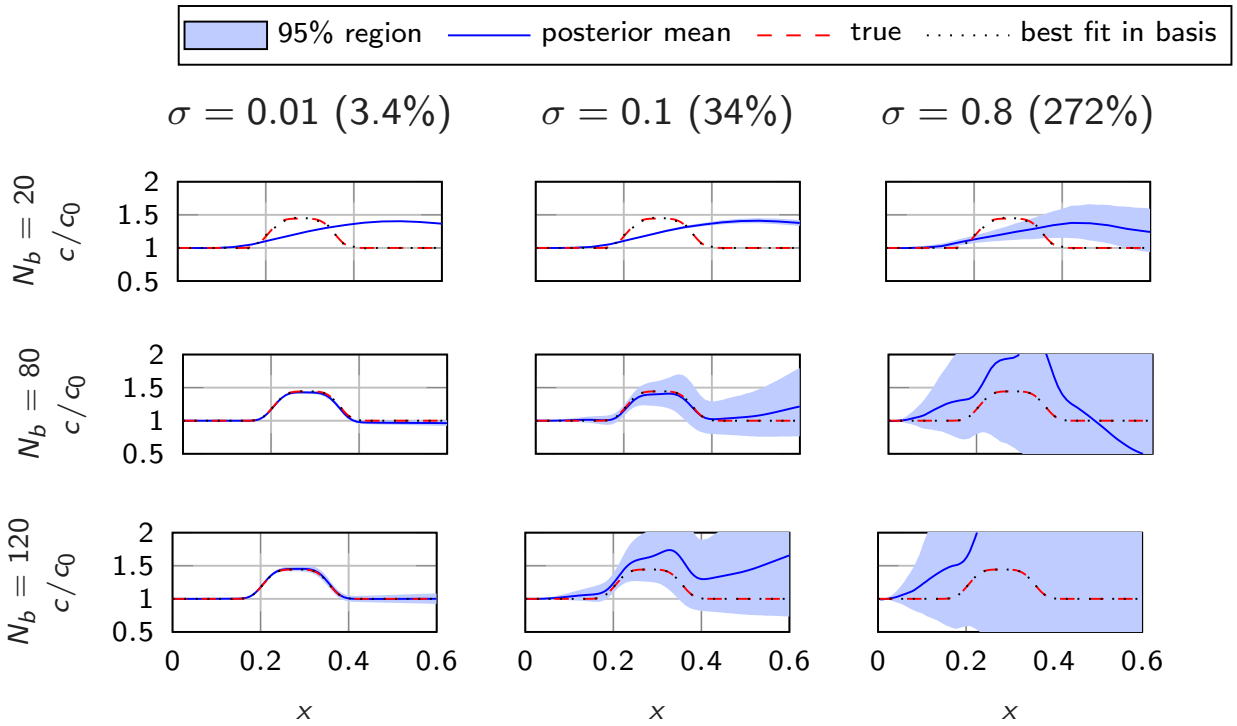
  A pure delay $g(t) = f(t-\tau)$ gives $W_2^2 = \tau^2$: the misfit grows smoothly with the shift
- Use $-W_2^2(f, g_\theta)/(2\varepsilon^2)$ in place of $\log L$: accounts for **transport** of the contrast when correcting the amplitude alone is too costly

Thank you!

Questions?

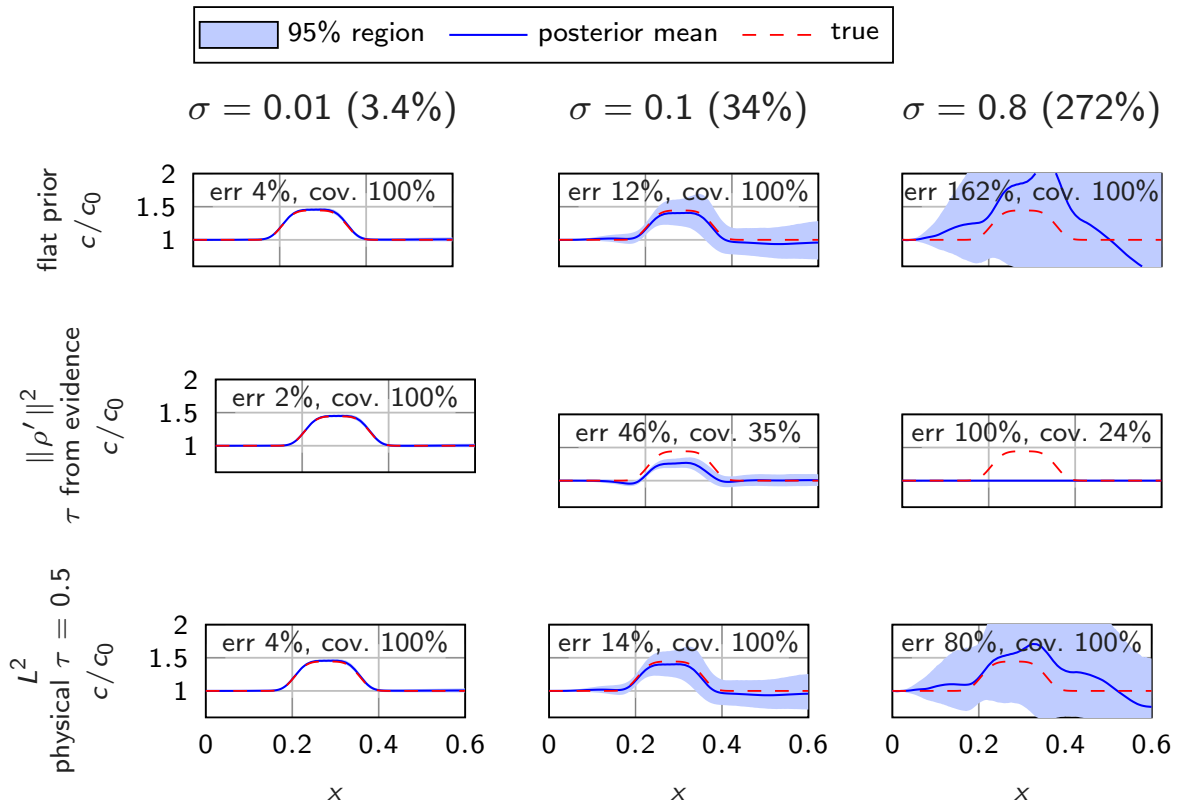
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Appendix A1: basis expansion with a flat prior



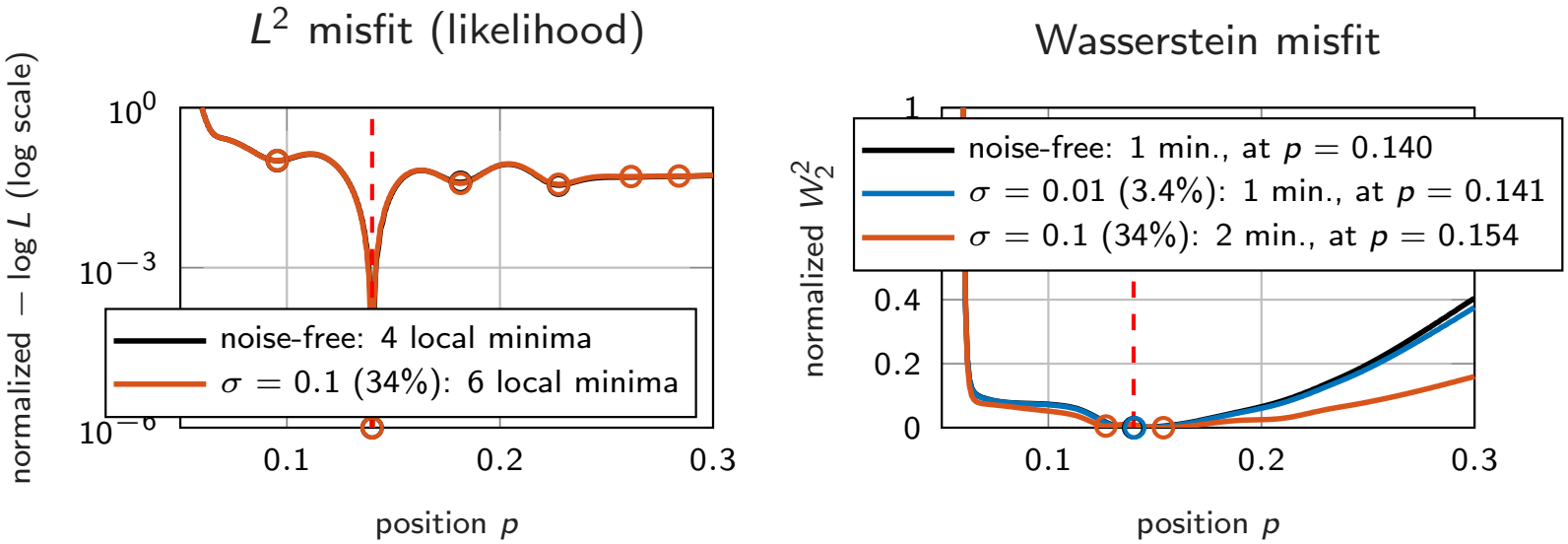
- $\rho = \sum_j \beta_j \phi_j$ (hat functions): linear in $\beta \Rightarrow$ Gaussian posterior in closed form; MCMC (full proposal covariance) reproduces it
- Spacing above $\lambda/2 \approx 0.017$ ($N_b \leq 40$): basis kinks excite the sensitive band, model error $\sim 100\times$ the noise $\Rightarrow$ **confidently wrong**
- Fine basis ($N_b = 120$): 2.4% error at 3.4% noise, but the variance explodes with noise. Gaussian basis: 4–5$\times$ smaller model error, works from $N_b = 80$

Appendix A2: Bayesian regularization on the basis



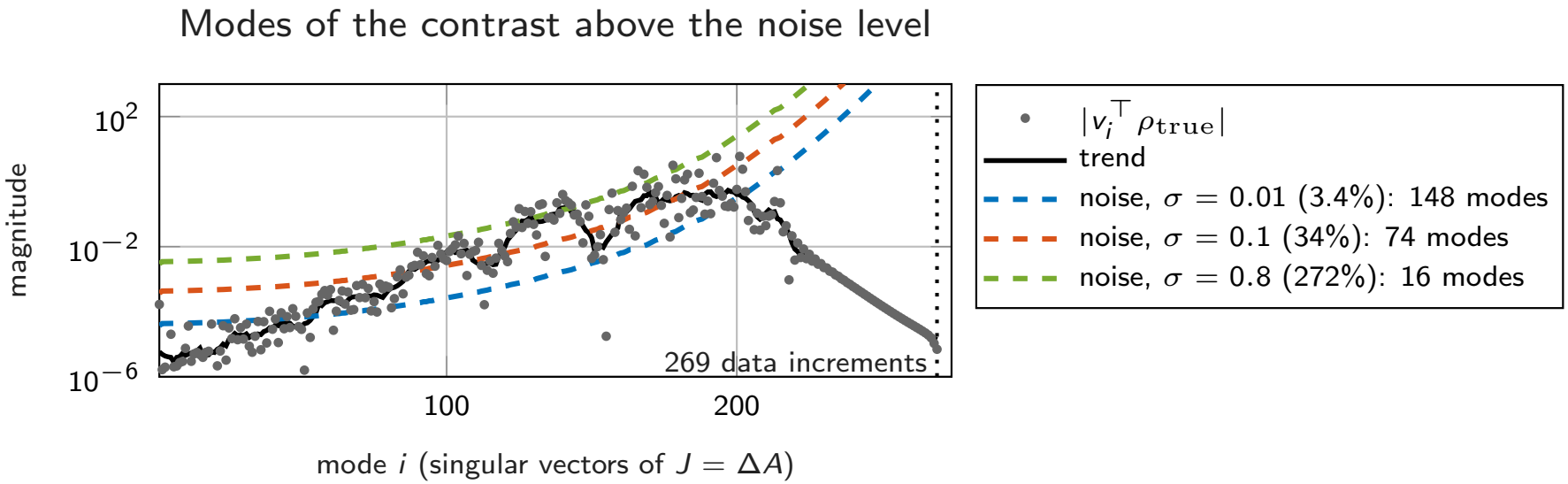
- Gaussian basis, $N_b = 80$. Prior $\propto e^{-\|\rho\|^2/(2\tau^2)}$ (L^2) or $e^{-\|\rho'\|^2/(2\tau^2)}$ (gradient): posterior still Gaussian, closed form
- τ from the evidence: best at low noise (2%), but it over-shrinks as the noise grows and **collapses to** $\rho = 0$ at 272% noise
- Physical τ: honest bands at every noise level; still $\sim 10\times$ less precise than the bump model $\Rightarrow$ the shape prior is the most informative. Next: hierarchical τ (Gibbs sampling)

Appendix A3: misfit landscape in the position, L^2 vs Wasserstein



- Along p (a, w fixed); W_2 between densities $e^{c\,s(t)}$ of the mean log-signal and the model signal
- L^2: 4–6 **local minima** (cycle skipping) trap far starts; the posterior stays unimodal (secondary minima $\sim e^{-2400}$)
- W_2: a **single well** (noise-free, 3.4%); at 34% noise the Brownian drift is transported too, biasing the minimum
- $\Rightarrow$ W_2 to find the basin, the L^2 likelihood (correct noise model) for the posterior and UQ

Appendix A4: how many modes can the data resolve?



- SVD of $J = \Delta A$ (row k: $A_{k+1,:} - A_{k,:}$): mode i has std $\sigma/(s_i\sqrt{M/\Delta t})$, resolvable if below $|v_i^\top \rho|$; at most 269 (data increments)
- Most sensitive modes oscillate at the reflection wavenumber $2/\lambda \approx 60$: the smooth bump is seen **through its edges**
- The data only reach $x \leq 0.83$ (round trip within T); smooth features there are fully visible, fine scales and deeper features are not

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